PlusMinus Analytics

What I set out to do

- I developed PlusMinus to connect basketball shooting results to a useful next step: deciding what to practice.
- The project brought together a historical NBA explorer and a separate training prototype, supported by product documentation.

How I organized the work

- I used a Product Requirements Document (PRD) to explain the problem, intended user, requirements, and acceptance criteria.
- The feature blueprint described each capability, and the user flows connected reviewing results to choosing a practice plan.
- The six-month roadmap outlined proposed development beyond the prototype. It was a planning horizon, not six months of completed delivery.

How I approached the data

- The historical sample contained 79,763 shots across 10 selected NBA players. The Python pipeline cleaned and aggregated those records for analysis.
- Training Lab used three fictional players, 24 sessions, and 1,440 synthetic shots so the demonstration could be repeated consistently.
- I kept those sources separate and made their limits clear. The NBA sample was not league-wide, and synthetic practice data could not prove real player improvement.

What the prototype demonstrated

- The model contained seven entities: players, sessions, shot maps, events, coaches, ratings, and plans.
- The player, coach, and training dashboards each contained five visuals, giving 15 visuals across three dashboards.
- Five simulated API contracts handled players, sessions, shot maps, ratings, and plans inside the browser. JSON powered the demo, with a separate SQLite artifact showing the database structure.

How the results became a practice plan

- The recommendation used the selected player's lowest-accuracy zone with at least 20 attempts, then divided the chosen time among warm-up, focused practice, and review.
- This made the plan specific to recorded results and available time. It remained a transparent rule, not live AI or a diagnosis of shooting technique.

PlusMinus Analytics: Decisions and lessons

Where data quality became important

- One concrete issue was the era classification, which could use a team abbreviation or misread a season identifier.
- The correction prioritized explicit season information, used dates as a fallback, and accounted for the 2020 bubble. Rebuilt exports and regression tests supported the repair.
- That example reinforced my main lesson: a chart can look reasonable while the underlying data means the wrong thing.

How I checked the numbers

- I defined overall accuracy as total makes divided by total attempts. Sessions of 1/2 and 9/10 combine to 10/12, or 83.3%, rather than an average of 70%.
- The validation checked record relationships, metric totals, plan durations, and simulated API responses. All four Node tests passed during the project review.
- I also distinguished descriptive measures from stronger claims. The shooting index was a session percentage, and the 20-attempt rule was not a statistical confidence threshold.

How I focused the next release

- I narrowed the proposed next release to logging a session, reviewing results, choosing a plan, and recording a comparable follow-up.
- The requirements covered reliable saving, duplicate prevention, and record ownership. I would measure whether users could finish unaided, understand the plan, and return.
- The production backend and user pilot remained proposed work. The goal was to evaluate the core workflow before adding more complex features.

How I explain my contribution and results

- I used AI coding assistance, so I distinguish the decisions I made from implementation I reviewed or validated. My ownership discussion centers on scope, definitions, and evidence.
- The case study brought together architecture, competitive analysis, proposed KPIs, and a go-to-market strategy. The implementation counts show scope; I cannot substantiate a threefold recruiter-engagement increase.

What I would bring to Vanguard

- This project helped me connect data discovery, clear business rules, validation, and communication. I would bring that same care to understanding and supporting trusted data products.